

# Top-k Ranking Preservation under Bounded Score Perturbations

Research Architecture Runtime

operator/top-k

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## Abstract

`top-k` is certified by pairwise ranking preservation when all gaps exceed  $2\varepsilon$ .  
Lean `LEAN_FULL` via `OrderStat.Ranking`.

**This theorem is:** Reduction

## 1 Problem

The `top-k` operator depends on the pairwise order of scores  $s \in \mathbb{R}^n$ . The certified object is the full pairwise ranking (and thereby any ranking-determined selection such as `top-k` index sets).

## 2 Stability notion

Perturbations:  $\|\delta\|_\infty \leq \varepsilon$ . Stability: if `AllGapsExceed`( $s, 2\varepsilon$ ), then  $s_i < s_j \Leftrightarrow (s + \delta)_i < (s + \delta)_j$  for all  $i, j$ . Sharpness: a gap  $\leq 2\varepsilon$  admits a collision adversary.

## 3 Definitions

**Definition 1** (All-gaps).  $|s_p - s_q| > g$  for  $p \neq q$ .

**Assumptions.**  $n \geq 2$ ,  $\varepsilon \geq 0$ , and (for invariance) all pairwise gaps  $> 2\varepsilon$ .

## 4 Theorem

**Theorem 2** (Ranking invariance). *If  $\text{AllGapsExceed}(s, 2\varepsilon)$ , then for every  $\|\delta\|_\infty \leq \varepsilon$  and all  $i, j$ ,  $s_i < s_j \Leftrightarrow s_i + \delta_i < s_j + \delta_j$ .*

**Theorem 3** (Sharpness). *If some distinct pair has  $|s_i - s_j| \leq 2\varepsilon$ , an admissible  $\delta$  forces a value collision.*

## 5 Intuition

Relative  $\ell_\infty$  moves cost at most  $2\varepsilon$ . Gaps larger than  $2\varepsilon$  cannot reverse; smaller gaps can be closed by a midpoint push.

## 6 Examples

**Example 4.** Scores  $(0, 3, 8)$ ,  $\varepsilon = 1$ : all gaps  $> 2$ , so the ranking is stable. If scores are  $(0, 1.5)$  and  $\varepsilon = 1$ , the gap  $1.5 \leq 2$  admits a midpoint tie.

## 7 Proof

Fix  $\|\delta\|_\infty \leq \varepsilon$ . If  $i = j$  both sides of the claimed equivalence are false for strict  $<$ . If  $i \neq j$  and  $|s_i - s_j| > 2\varepsilon$ , the estimate  $(s_i + \delta_i) - (s_j + \delta_j) \geq (s_i - s_j) - 2\varepsilon$  (and its swap) yields  $s_i < s_j \Leftrightarrow s_i + \delta_i < s_j + \delta_j$  (Theorem 2).

For sharpness, take  $i \neq j$  with  $|s_i - s_j| \leq 2\varepsilon$  and the midpoint  $\delta_i = -(s_i - s_j)/2$ ,  $\delta_j = (s_i - s_j)/2$  (elsewhere 0). Then  $\|\delta\|_\infty \leq \varepsilon$  and  $s_i + \delta_i = s_j + \delta_j$  (Theorem 3).

## 8 Formal statement

Lean module

`Research.Operators.TopK.Preservation`

Source file

`lean/Research/Operators/TopK/Preservation.lean`

Theorems

- `top\_k\_margin\_invariance`
- `top\_k\_margin\_sharpness`

Note

Reduction of `OrderStat.Ranking`.

## Certificate

`lean/certificates/top-k/top-k-margin`

## Status

LEAN\_FULL on Mathlib  $\mathbb{R}$  (REAL\_MATHLIB)

# 9 Proof dependencies

- `Research.Operators.OrderStat.Ranking` / shared pairwise lemma from `KthMargin`.
- Midpoint collision adversary.
- No other operator theorems required.

# 10 Consequences

Ranking stability implies stability of any functional of the pairwise order used by `top-k`. Related operators: `sorting`, `rank`, `kth-order-statistic`.